Experiment No. 5

Title

Text Pattern Searching and Log Analysis in Linux Using grep and Related Utilities

Aim

To understand and implement Linux text-searching commands such as grep, fgrep, and related options in order to efficiently locate specific patterns, keywords, and strings within files and command outputs for effective log analysis, troubleshooting, and automation tasks.

Theory

Text Searching in Linux

Linux treats most data as plain text, making text-processing tools extremely powerful. Searching text efficiently is essential for:

- Log file analysis
- Configuration troubleshooting
- Security monitoring
- Automation scripts

grep Command

The grep command searches for a pattern in files or input streams and prints matching lines. Basic syntax:

grep [options] pattern filename

Related Commands

- grep – General pattern searching
- fgrep – Fast grep (searches fixed strings, no regular expressions)
- egrep – Extended grep (supports extended regular expressions)

Note: In modern Linux systems, egrep and fgrep are implemented using grep -E and grep -F.

Commands Used

Commands	Description
grep	Search text using patterns
grep -i	Case-insensitive search
grep -n	Show line numbers
grep -r	Recursive search
grep -v	Invert match
fgrep /grep -F	Search fixed strings
cat	Display file content
ls	List files

Procedure / Steps

Step 1: Login

- Log in to the Linux virtual machine and open the **Terminal**.

Step 2: Create a working directory:

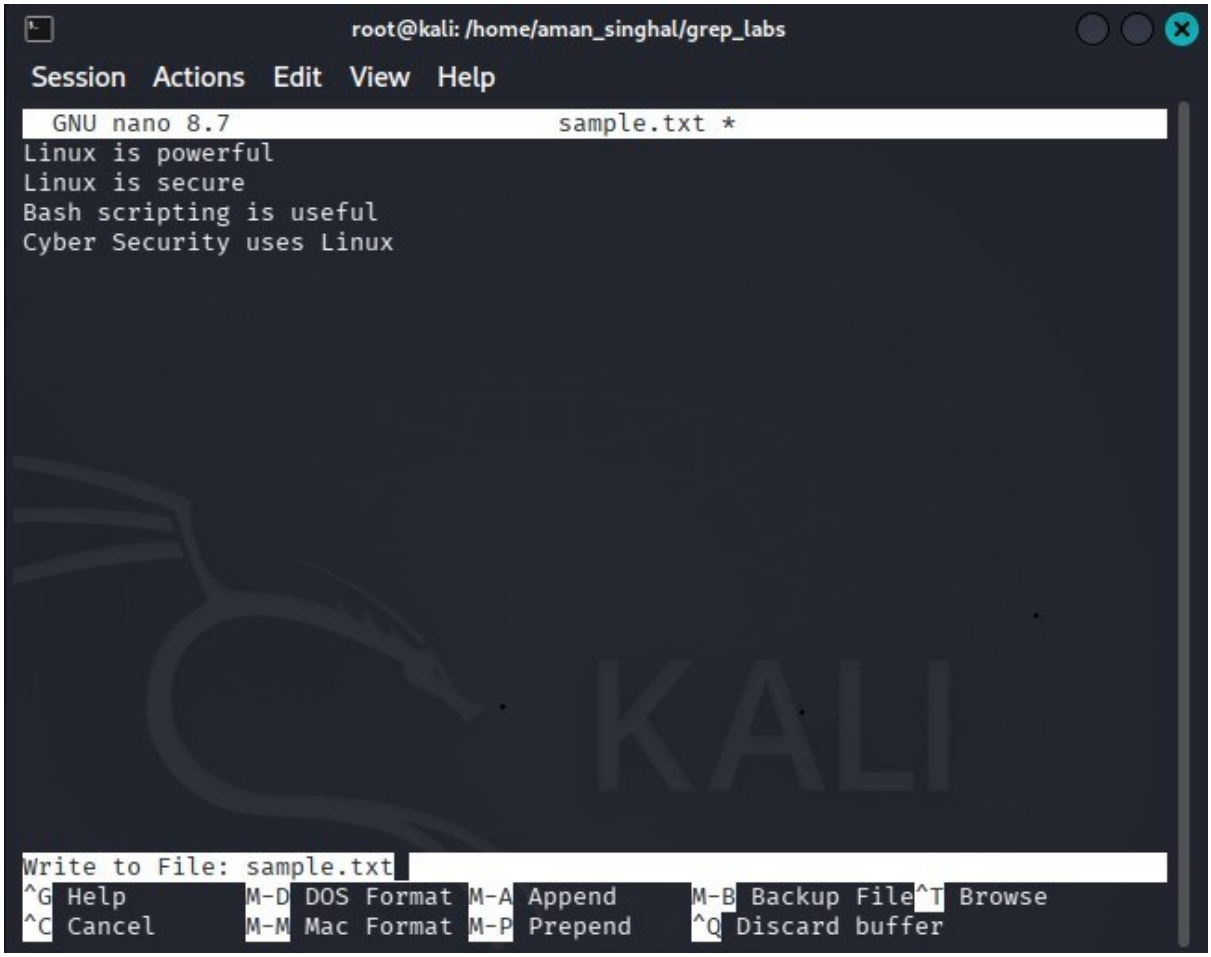
- mkdir grep_lab
- cd grep_lab

Step 3: Create a sample text file:

- nano sample.txt

Step 4: Add content to files using redirection:

- Linux is powerful
- Linux is secure
- Bash scripting is useful
- Cyber Security uses Linux



Step 5: Display file contents:

- cat file1.txt

```
root@kali: /home/aman_singhal/grep_labs
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal/grep_labs]
└─# cat file1.txt
Linux is powerful Linux is secure Bash scripting is useful Cyber
Security uses Linux
Linux is powerful
Linux is secure
Bash scripting is useful
Cyber Security uses Linux

┌──(root㉿kali)-[/home/aman_singhal/grep_labs]
└─#
```

Step 6: Search for a word using grep:

- grep Linux sample.txt

```
root@kali: /home/aman_singhal/grep_labs
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal/grep_labs]
└─# grep linux sample.txt

┌──(root㉿kali)-[/home/aman_singhal/grep_labs]
└─#
```

Step 7: Perform case-insensitive search:

- grep -i linux sample.txt

```
root@kali: /home/aman_singhal/grep_labs
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal/grep_labs]
└─# grep -i linux sample.txt
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┌──(root㉿kali)-[/home/aman_singhal/grep_labs]
└─#
```

Step 8: Display line numbers with matches:

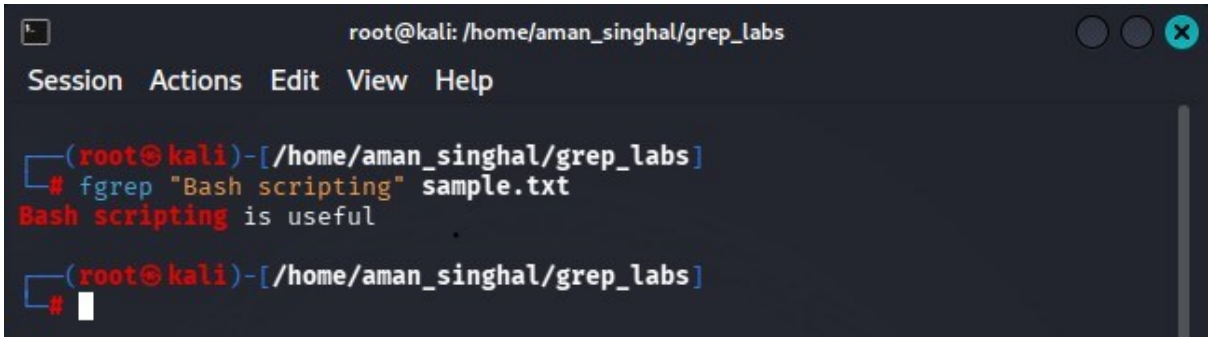
- grep -n Linux sample.txt

Step 9: Invert the match:

- grep -v Linux sample.txt

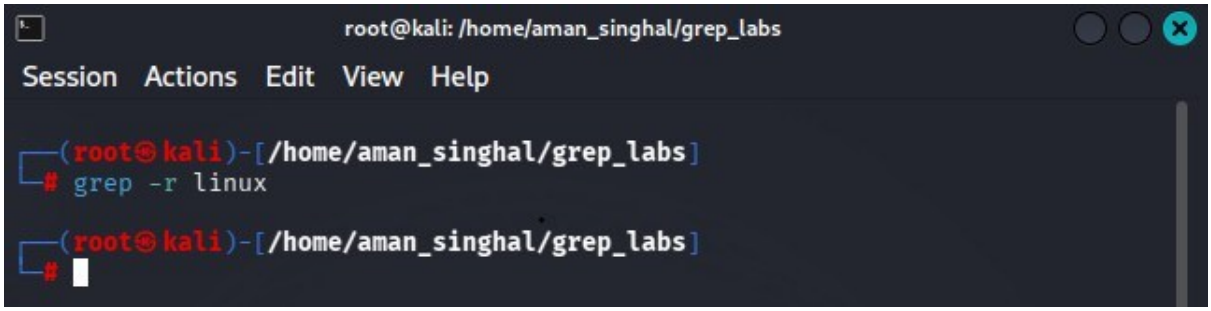
Step 10: Search using fgrep:

- fgrep "Bash scripting" sample.txt



Step 11: Perform recursive search (if multiple files exist):

- grep -r Linux



Outcomes

- Use grep, fgrep, and related commands to search for specific patterns in text files.
- Apply various search options such as case-insensitive search (-i), recursive search (-r), inverted match (-v), line numbers (-n), and count (-c).
- Analyze log files to identify errors, warnings, and important system events.
- Combine search commands with other Linux utilities using pipes (|) for advanced filtering.
- Automate pattern-search tasks in scripts for efficient system administration.
- Understand the role of text-searching tools in troubleshooting and cybersecurity monitoring.

Result

- Lines matching the specified search pattern are correctly displayed.
- Line numbers are shown when the appropriate option is used.
- Case-insensitive searches return accurate results.
- Inverted searches successfully display non-matching lines.
- Recursive searches correctly scan directories and subdirectories.
- The count option accurately displays the number of matching lines.